FCFS Scheduling

```
#include <stdio.h>

void WaitingTime(int process_ids[], int num_processes,int burst_times[],
int waiting_times[])
{
    waiting_times[0] = 0;
    for (int i = 1; i < num_processes; i++)
        waiting_times[i] = burst_times[i - 1] + waiting_times[i - 1];
}


void TurnAroundTime(int process_ids[], int num_processes, int
burst_times[], int waiting_times[], int turn_around_times[])
{
    for (int i = 0; i < num_processes; i++)
    {
        turn_around_times[i] = burst_times[i] + waiting_times[i];
    }
}


void avgTime(int process_ids[], int num_processes, int burst_times[])
{
    int waiting_times[num_processes], turn_around_times[num_processes];
    int total_waiting_time = 0, total_turn_around_time = 0;


    WaitingTime(process_ids, num_processes, burst_times, waiting_times);
    TurnAroundTime(process_ids, num_processes, burst_times,
waiting_times, turn_around_times);


    printf("Processes   Burst time   Waiting time   Turn around time\n");


    for (int i = 0; i < num_processes; i++)
    {
        total_waiting_time += waiting_times[i];
```

```
        total_turn_around_time += turn_around_times[i];

        printf("\t%d ", (i + 1));

        printf("\t%d ", burst_times[i]);

        printf("\t%d", waiting_times[i]);

        printf("\t\t%d\n", turn_around_times[i]);

    }

    int avg_waiting_time = (float)total_waiting_time /
(float)num_processes;

    int avg_turn_around_time = (float)total_turn_around_time /
(float)num_processes;

    printf("Average waiting time = %d", avg_waiting_time);

    printf("\n");

    printf("Average turn around time = %d ", avg_turn_around_time);

}


int main()

{

    int process_ids[] = {1, 2, 3};

    int num_processes = sizeof process_ids / sizeof process_ids[0];


    int burst_time[] = {5, 6, 2};

    avgTime(process_ids, num_processes, burst_time);

    return 0;

}
```

Output:

```
[Running] cd "g:\OS\lab4\" && gcc fcfs.c -o fcfs && "g:\OS\lab4\"fcfs
Processes    Burst time   Waiting time   Turn around time
    1          5            0              5
    2          6            5              11
    3          2            11             13
Average waiting time = 5
Average turn around time = 9
[Done] exited with code=0 in 0.724 seconds
```

SJF Scheduling

```
 #include<stdio.h>
 void main()
{
    int bt[20],p[20],wt[20],tat[20],i,j,n,total=0,pos,temp;  //bt-Burst
Time // wt-Waiting Time  //  tat-Turn Around Time  //pos-position
    float avg_wt,avg_tat;
    printf("Enter number of process:");
    scanf("%d",&n);
    printf("\nEnter Burst Time:\n");
    for(i=0;i<n;i++)
    {
        printf("p%d:",i+1);
        scanf("%d",&bt[i]);
        p[i]=i+1;           //contains process number
    }

    //sorting burst time in ascending order
    for(i=0;i<n;i++)
    {
        pos=i;
        for(j=i+1;j<n;j++)
        {
            if(bt[j]<bt[pos])
                pos=j;
        }

        temp=bt[i];
        bt[i]=bt[pos];
        bt[pos]=temp;
        temp=p[i];
```

```
        p[i]=p[pos];
        p[pos]=temp;
    }

    wt[0]=0;            //waiting time for first process will be zero

    //calculate waiting time
    for(i=1;i<n;i++)
    {
        wt[i]=0;
        for(j=0;j<i;j++)
         wt[i]+=bt[j];
        total+=wt[i];
    }
    avg_wt=(float)total/n;      //average waiting time
    total=0;

    printf("\nProcess\t    Burst Time    \tWaiting Time\tTurnaround
Time");
    for(i=0;i<n;i++)
    {
        tat[i]=bt[i]+wt[i];     //calculate turnaround time
        total+=tat[i];
        printf("\np%d\t\t  %d\t\t
%d\t\t\t%d",p[i],bt[i],wt[i],tat[i]);
    }
    avg_tat=(float)total/n;     //average turnaround time
    printf("\n\nAverage Waiting Time=%f",avg_wt);
    printf("\nAverage Turnaround Time=%f\n",avg_tat);
}
```

Output:

```
G:\OS\lab4>sjf.exe
Enter number of process:3

Enter Burst Time:
p1:1
p2:2
p3:3

Process       Burst Time              Waiting Time     Turnaround Time
p1                 1                      0                   1
p2                 2                      1                   3
p3                 3                      3                   6

Average Waiting Time=1.333333
Average Turnaround Time=3.333333
```

Priority Scheduling

```
#include <stdio.h>
void swap(int *a,int *b)
{
    int temp=*a;
    *a=*b;
    *b=temp;
}
int main()
{
    int n;
    printf("Enter Number of Processes: ");
    scanf("%d",&n);

    int burst[n],priority[n],index[n];
    for(int i=0;i<n;i++)
    {
        printf("Enter Burst Time and Priority Value for Process %d:
",i+1);
        scanf("%d %d",&burst[i],&priority[i]);
        index[i]=i+1;
    }
    for(int i=0;i<n;i++)
    {
        int temp=priority[i],m=i;

        for(int j=i;j<n;j++)
        {
            if(priority[j] > temp)
            {
                temp=priority[j];
                m=j;
```

```
            }
        }

        swap(&priority[i], &priority[m]);
        swap(&burst[i], &burst[m]);
        swap(&index[i],&index[m]);
    }

    int t=0;

    printf("Order of process Execution is\n");
    for(int i=0;i<n;i++)
    {
        printf("P%d is executed from %d to %d\n",index[i],t,t+burst[i]);
        t+=burst[i];
    }
    printf("\n");
    printf("Process Id\tBurst Time\tWait Time\n");
    int wait_time=0;
    int total_wait_time = 0;
    for(int i=0;i<n;i++)
    {
        printf("P%d\t\t%d\t\t%d\n",index[i],burst[i],wait_time);
        total_wait_time += wait_time;
        wait_time += burst[i];
    }

    float avg_wait_time = (float) total_wait_time / n;
    printf("Average waiting time is %f\n", avg_wait_time);

    int total_Turn_Around = 0;
    for(int i=0; i < n; i++){
```

```
        total_Turn_Around += burst[i];
    }
    float avg_Turn_Around = (float) total_Turn_Around / n;
    printf("Average TurnAround Time is %f",avg_Turn_Around);
    return 0;
}
```

Output:

```
G:\OS\lab4>gcc priorityScheduling.c -o ps.exe

G:\OS\lab4>ps.exe
Enter Number of Processes: 3
Enter Burst Time and Priority Value for Process 1: 1 1
Enter Burst Time and Priority Value for Process 2: 2 2
Enter Burst Time and Priority Value for Process 3: 3 3
Order of process Execution is
P3 is executed from 0 to 3
P2 is executed from 3 to 5
P1 is executed from 5 to 6

Process Id      Burst Time      Wait Time
P3              3               0
P2              2               3
P1              1               5
Average waiting time is 2.666667
Average TurnAround Time is 2.000000
```

Round Robins Scheduling

```
#include<stdio.h>

int main()

{

 int cnt,j,n,t,remain,flag=0,tq;

 int wt=0,tat=0,at[10],bt[10],rt[10];

 printf("Enter Total Process:\t ");

 scanf("%d",&n);

 remain=n;

 for(cnt=0;cnt<n;cnt++)

 {

  printf("Enter Arrival Time and Burst Time for Process Process Number %d
:",cnt+1);

  scanf("%d",&at[cnt]);

  scanf("%d",&bt[cnt]);

  rt[cnt]=bt[cnt];

 }

 printf("Enter Time Quantum:\t");

 scanf("%d",&tq);

 printf("\n\nProcess\t|Turnaround Time|Waiting Time\n\n");

 for(t=0,cnt=0;remain!=0;)

 {

  if(rt[cnt]<=tq && rt[cnt]>0)

  {

   t+=rt[cnt];

   rt[cnt]=0;

   flag=1;
```

```
   }
   else if(rt[cnt]>0)
   {
    rt[cnt]-=tq;
    t+=tq;
   }
   if(rt[cnt]==0 && flag==1)
   {
    remain--;
    printf("P[%d]\t|\t%d\t|\t%d\n",cnt+1,t-at[cnt],t-at[cnt]-bt[cnt]);
    wt+=t-at[cnt]-bt[cnt];
    tat+=t-at[cnt];
    flag=0;
   }
   if(cnt==n-1)
    cnt=0;
   else if(at[cnt+1]<=t)
    cnt++;
   else
    cnt=0;
  }
  printf("\nAverage Waiting Time= %f\n",wt*1.0/n);
  printf("Avg Turnaround Time = %f",tat*1.0/n);


  return 0;
}
```

Output:

```
G:\OS\lab4>rr.exe
Enter Total Process:     3
Enter Arrival Time and Burst Time for Process Process Number 1 :0 1
Enter Arrival Time and Burst Time for Process Process Number 2 :1 2
Enter Arrival Time and Burst Time for Process Process Number 3 :2 3
Enter Time Quantum:     1


Process |Turnaround Time|Waiting Time

P[1]    |       1       |       0
P[2]    |       3       |       1
P[3]    |       4       |       1

Average Waiting Time= 0.666667
Avg Turnaround Time = 2.666667
```